SOME ADDITIONAL MATHEMATICAL ACTIVITIES IN MATHEMATICAL EDUCATION IN LITHUANIA

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There is no doubt about it that the solving of mathematical problems is engaging, useful and creative but not very easy and sometimes even rather difficult task. The main question and most important task in that direction is what could to be done to make the maximal number of people to feel and strongly believe that it is worth doing, worth undertaking efforts of every kind and is extremally useful for the development of the deepness of mind.?

An attempts in that direction in being taken since many years in many countries among them also in Lithuania. After all mathematics as a subject suits perhaps most perfectly to make every engaged person to think deeper and to be able understand more than before.

It is clear that such a talk ought to begin from matters around the school and teaching of maths in it or, generally speaking, from the education system. Afterwards we are forced to speak what could be done in order to improve the level of students who have to deal with the average school standards. Matters of such a kind were not very actual in former days when only a very bounded number of persons were able to get an education, say, hundred or two hundred yers ago. In our formally very democratical time there aroused the real problem how to give a suitable mathematical education or how to learn the students think under the circumstances when they are to forced to study more subjects spending for each less time.

Because we are starting first of all from the Lithuanian background, let us make a short historical discourse. On the other hand, the experience shows that these problems are similar in the whole world. The main problem, task and challenge is how to make the process of problem solving attractive?

It ought to be told that in the Soviet Union the system of education was a strictly authoritarian system with the only announced and for everybody known possible truth and with the corresponding education. In the same time such education system served rather well for spreading the concrete information among others and mayby first of in the area of exact sciences. This could serve also as a possible explaination for the Sputnik and related things.

In Lithuania (in that period being a part of the Soviet Union as well as in some other so-called Union Republics) the mathematical Olympiads for high school students started in the year 1951. The first Olympiad in Lithuania was being organized under guidance of Jonas Kubilius This very first Olympiad organized in Lithuania was organized as Vilnius - the capital city of Lithuania - Olympiad but all the following mathematical Olympiads were already whole Lithuanian Olympiads.

By the way in that period the specialized mathematical schools on the base and by support of Universities were organized in many Republics of the former Soviet Union, first of all in the leading centres such as Moscow, Sankt-Petersburg (former Leningrad), Kiev, Novosibirsk. From that point of view Lithuania unfortunately was an exception.

In the year 1986 the idea to organize the mathematical team-contest in Lithuania was practically carried into life by Algirdas Zabulionis and this contest was named after already mentioned meanwhile well-known mathematician prof. Jonas Kubilius, who was also for more than 30 years a rector of the Vilnius University.

The idea was to gather the students and to propose them to work "in corpore" as if they were one person. The invitation of teams is based on the regional principle - that means if the corresponding region has some achievements in the "usual" individual Olympiad then they are invited to participate. The team consist from 5 persons and is working in the separate room for 4 hours. Immediately after that the estimation of their papers follows and is provided under guidance of the mathematicians mainly from the Department of Mathematics and Informatics of Vilnius University and afterwards in the very same day the winners are announced.

This Olympiad proved itself rather useful in many aspects.

First of all, in the period of regaining the Indepedence of Lithuania and other (but not only) Baltic states the Lithuanian team-contest gave perhaps the main impuls – thanks the activities of Latvian professor Agnis Andžans as well as by already mentioned Algirdas Zabulionis - to the team-contest of Baltic region named “Baltic Way” in order to honour the chain of peoples connecting hand by hand all three capitals of Baltic republics from Vilnius via Riga till Tallinn (600 km).

In first phase in that already international team-contest only Baltic states (Estonia, Latvia und Lithuania) participated. The next country which joined this team-contest was Finland and afterwards by other countries of Baltic region and also by Iceland which is “superBaltic country” because iceland was a first country which recognized the Independence of Baltic states.

In this context we in Lithuania are very proud to mention that in the Millennium year 2000 the Lithuanian team-contest was opened by prof. Peter Taylor who exactly in these days visited Lithuania.

Another offspring of the Lithuanian team-contest was the individual Lithuanian Olympiad for youngsters . It serves as trainings base for the invited teams and for engagemant of youngsters. In this year 2005 we will enjoy already the 7th edition of that Olympiad which is meanwhile provided in two levels – one for the grades 5-6 and another for the grades 7-8.

In this period the net of regional Olympiads in many places of Lithuania are organized Some of them are individual, some are organized as team contests and some are mixed.

The important thing in the non-formal high-school mathematical education was also the Lithuanian school for high school students. The first period of that voluntary two-years school for the high-school students lasted till 1989 – that means in the Soviet time. This school was provided by the mathematicians of the Vilnius University. For the students of that school some theoretical materials were presented, control works proposed and the answers marked. Afterwards this school ceased its activities and was renewed in the year 1998 on the wider basis. The leading force in this renewal was dr. Antanas Apynis, who also became the head of the Advisery Board of that School. The deputy director is the first author of that contribution.

Now the time of that distant-education lasts also two years, the students pay for this 2-years educations extremally symbolical fee of approximately 7 Euro, they are expected to solve 8 series of problems with 10 problems in each. All these problems they may find in corresponding Internet site. as well as the entering test. Then if the entering test is done succesfully each student gets its own password and at the very end of education these who solved sufficiently well are invited to come to Vilnius for the final test consisting of 4 problems for the solving of which 1 hour is given.

After reasonable result also in the final test the participants get the special certification which is the same time also a recommendation for the studying of exact sciences and give some privileges by entering the exact science studies in some Lithuanian universities All materials concerning the school are given in the Internet site http://www.mif.vu.lt/ljmm. In the next future this site will also provide the English version with most important information and with the texts of all proposed problems.

The corresponding materials – all proposed problems with the exact solutions - after every cycle of two years are published so that these are acceptable for everyone who have interest in it.

After the renewal in 6 issues approximately 2000 students got the certifications that they succeded in that school. Also 6 books (each containing more than 100 pages) were published.

The important thing for non-formal mathematical education is also to possess an approach to the media where it could be possible to present, discuss and show how to solve some nice, simple and useful mathematical problem. In Lithuania the second author enjoys such a possibility since 1998 writing a series of articles on these subjects in the magazin for informatics (Computerics and PC magazin)

It is also remarkable that since 3 years the number of participants of Cangaroo competition in Lithuania is about 60 000 (the population of Lithuania is under 4 Mio).

These activities are not easy but we have no choice if we are really taking care about our mathematival future.